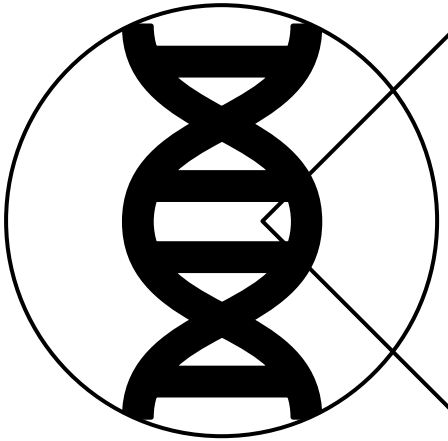
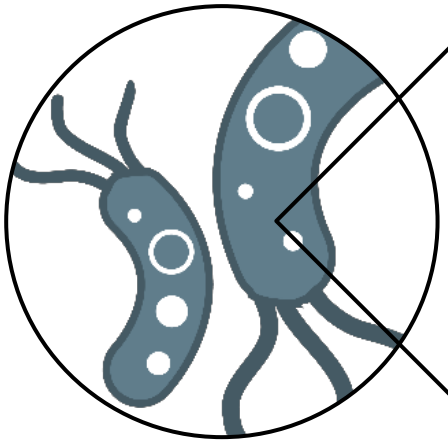
Objectives



Pathogenesis and mechanisms of enteric bacterial infections



Properties and characteristics of common and important pathogens that cause enteric bacterial infections






www.4allpetlovers.com




www.ColonixReview.com



Major syndromes of enteric infection

non-inflammatory watery



Gastroenteritis

No fever or low grade fever

No white blood cells in stool sample

- *Vibrio cholerae*
- ETEC
- EAggEC
- EPEC

Secretory diarrhoea

Fever

White blood cells in stool sample (neutrophils)

- *Campylobacter jejuni*
- *Shigella* spp.
- Non-typhoidal *Salmonella* serotypes
- EIEC
- EHEC

Inflammatory diarrhoea



Febrile patient

Fever

White blood cells in stool sample (mononuclear cells)

- Typhoidal *Salmonella* serotypes
- Enteropathogenic *Yersinia* spp.
- *Brucella* spp.

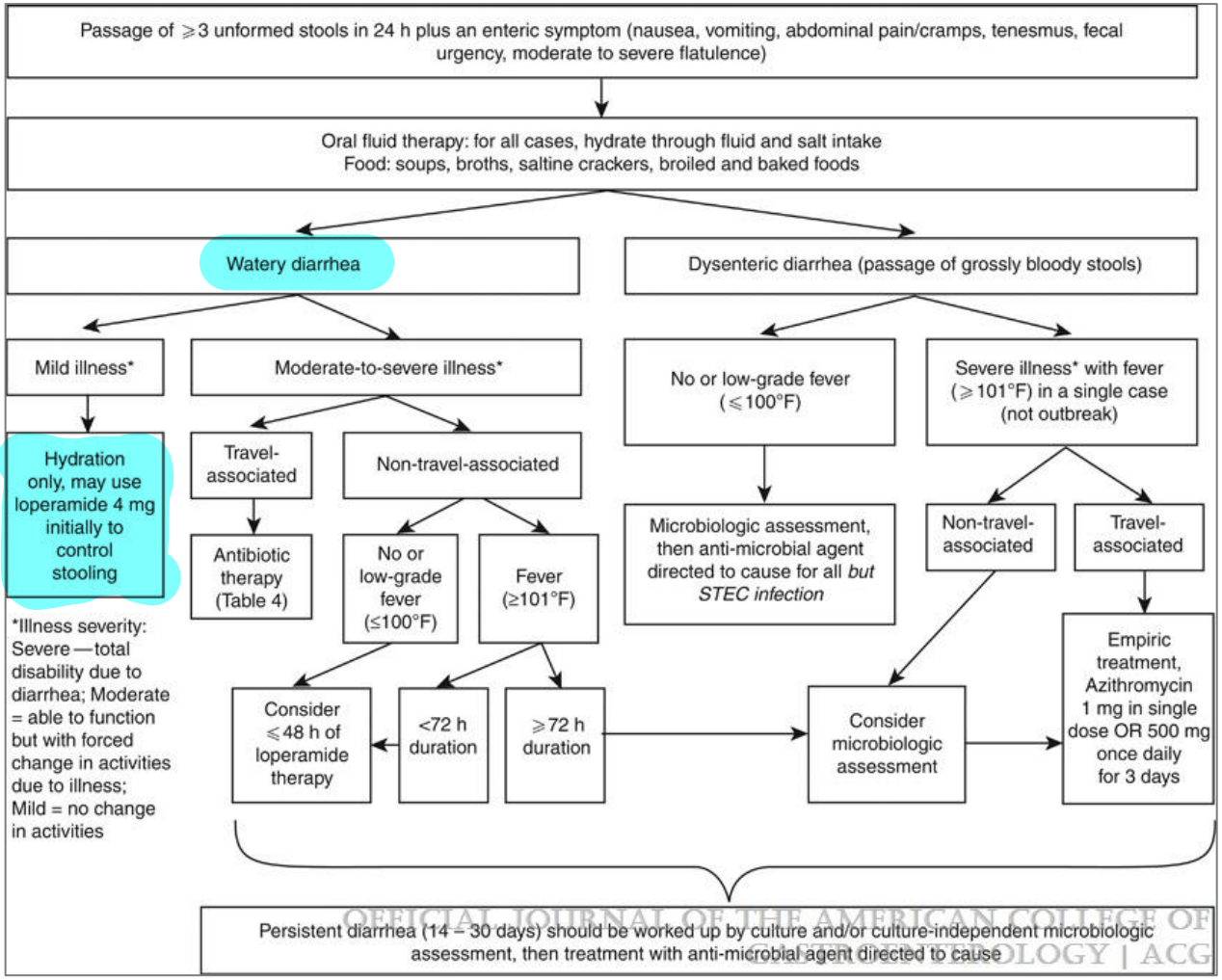
Enteric fever

Nat Rev Microbiol 2008

Why do we need to categorize type of diarrhea?

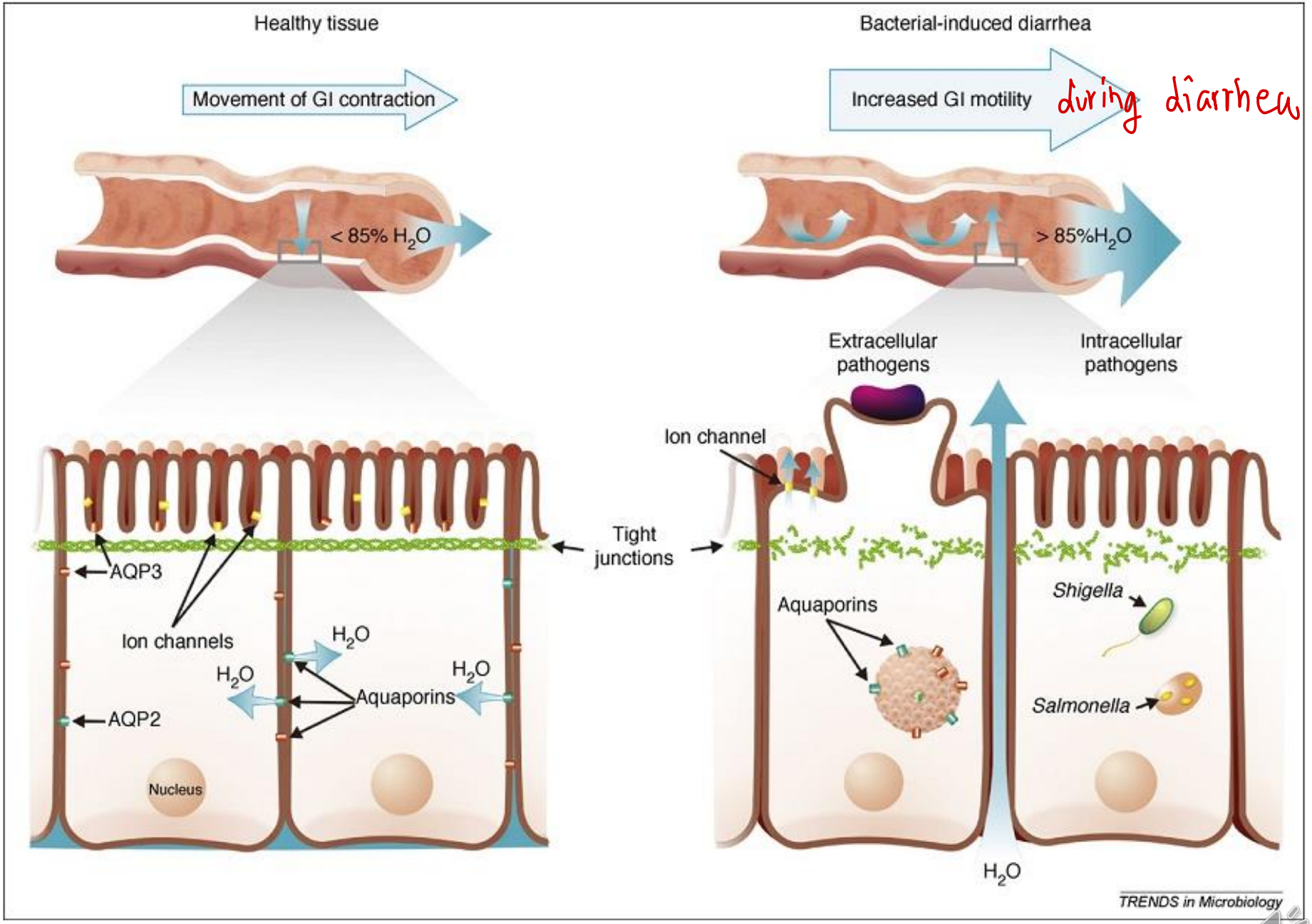


Management of adult patients with acute diarrhea (suspect infectious etiology)

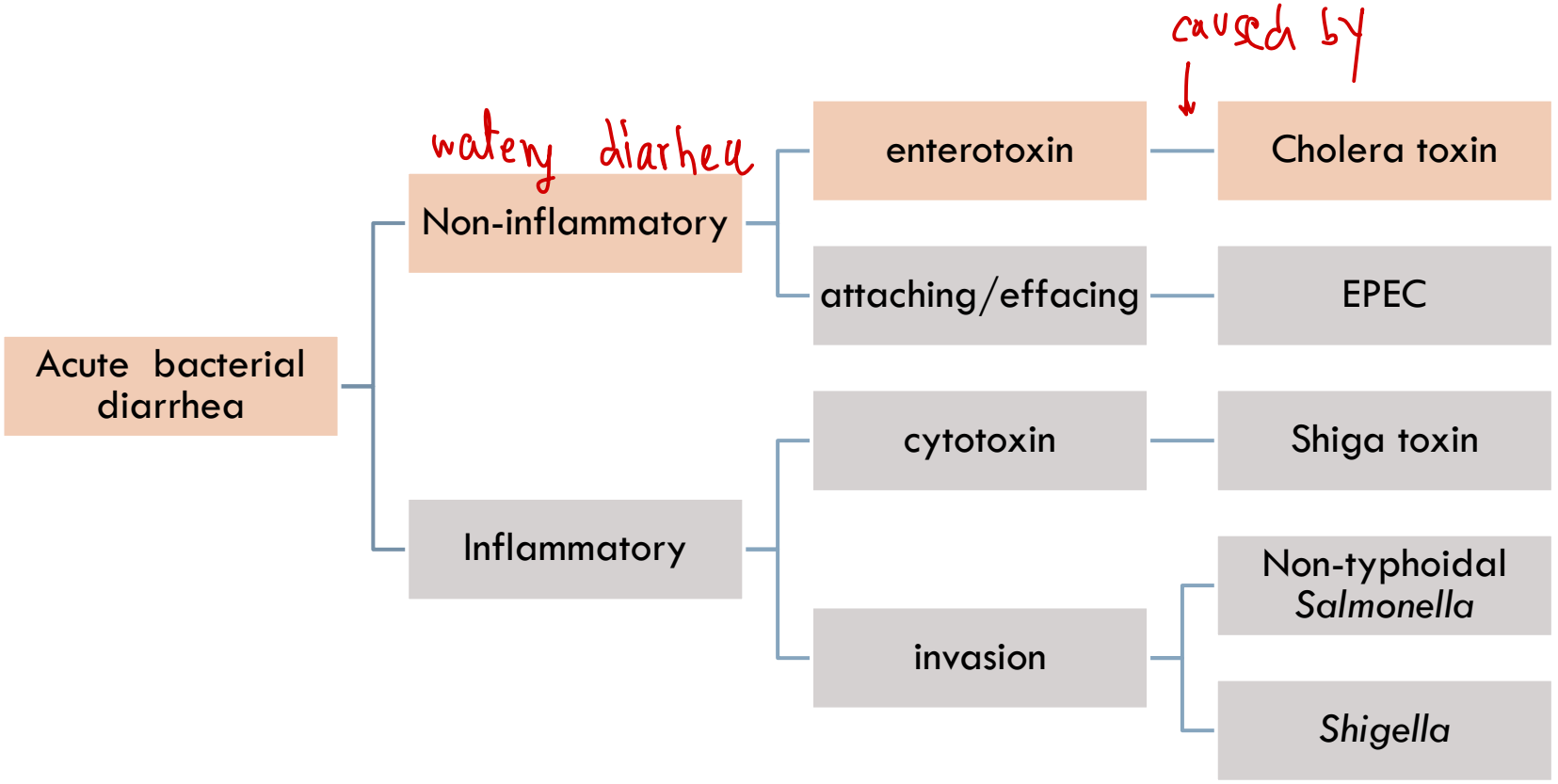


journal of the American College of Gastroenterology | ACG111(5):602-622, May 2016. doi: 10.1038/ajg.2016.126

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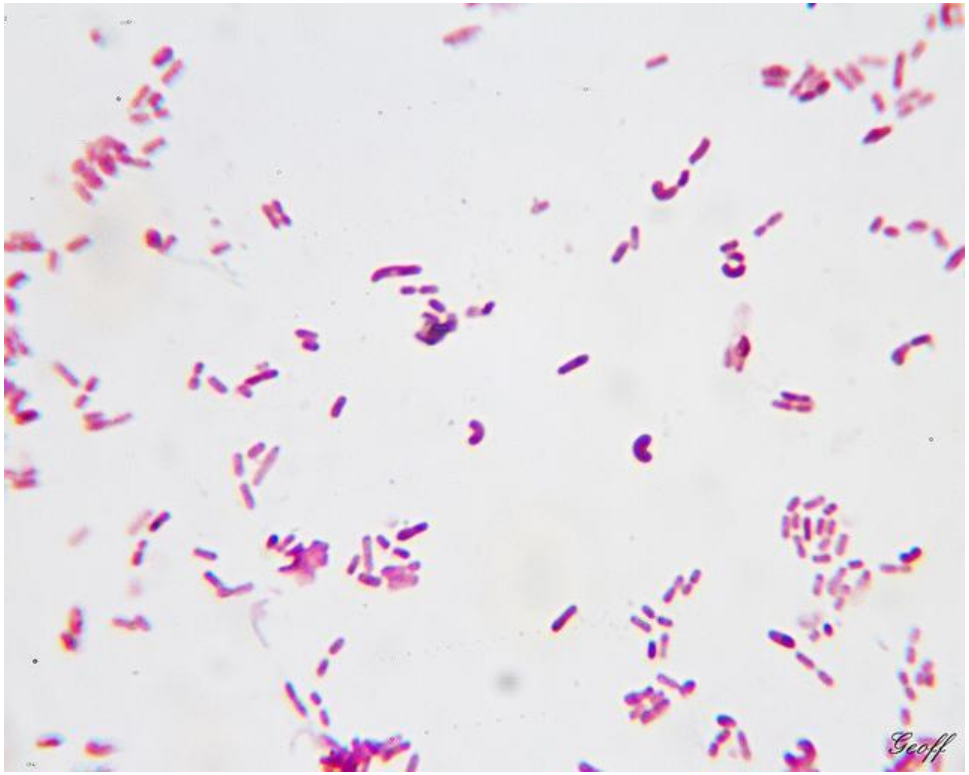

Healthy tissue
Movement of GI contraction
< 85% H_2O
AQP3
Ion channels
H_2O
H_2O
H_2O
Aquaporins
AQP2
Nucleus
Tight junctions
Bacterial-induced diarrhea
Increased GI motility
during diarrhea
> 85%H_2O
Extracellular pathogens
Intracellular pathogens
Ion channel
Aquaporins
Shigella
Salmonella
H_2O
TRENDS in Microbiology

Pathogenesis of acute bacterial diarrhea



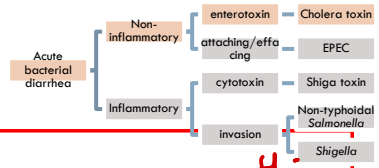
Vibrio spp.

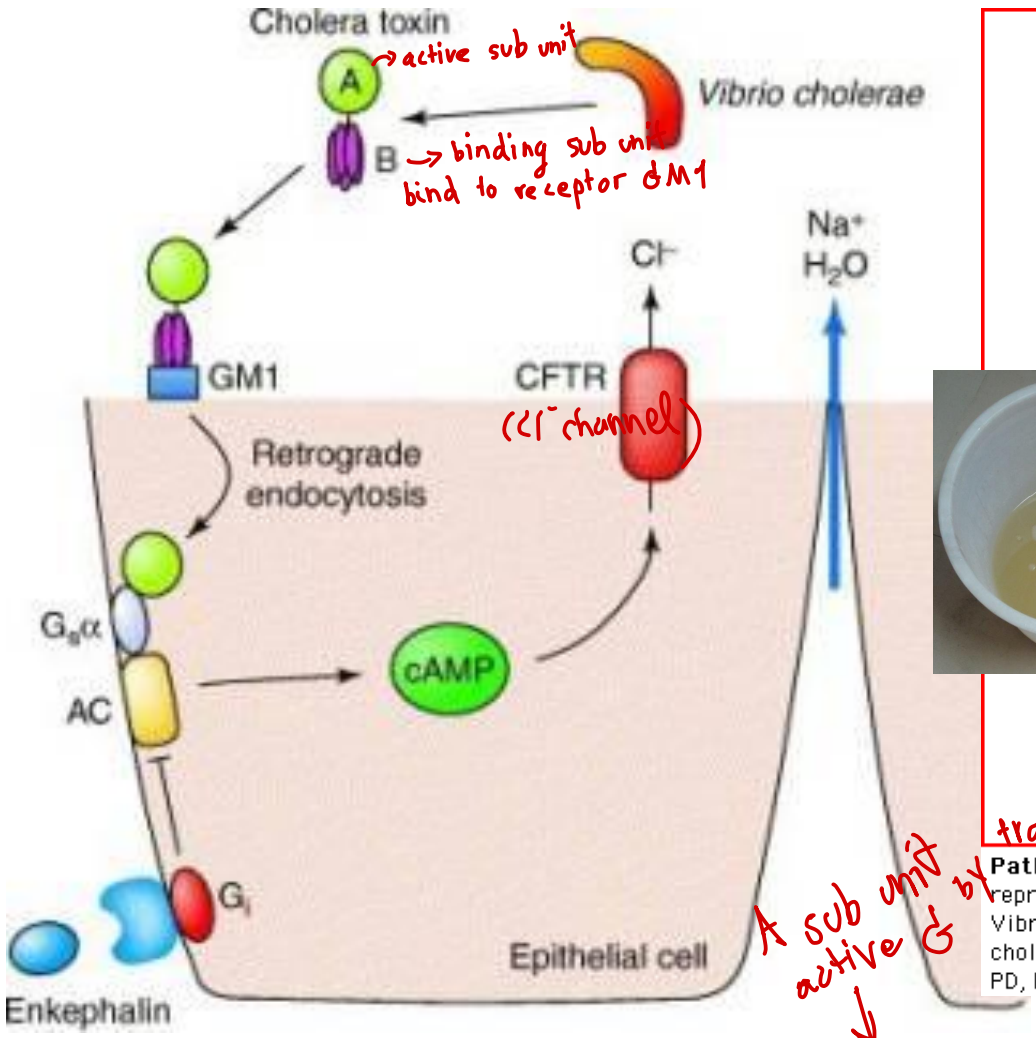
secrete -> cholera toxin

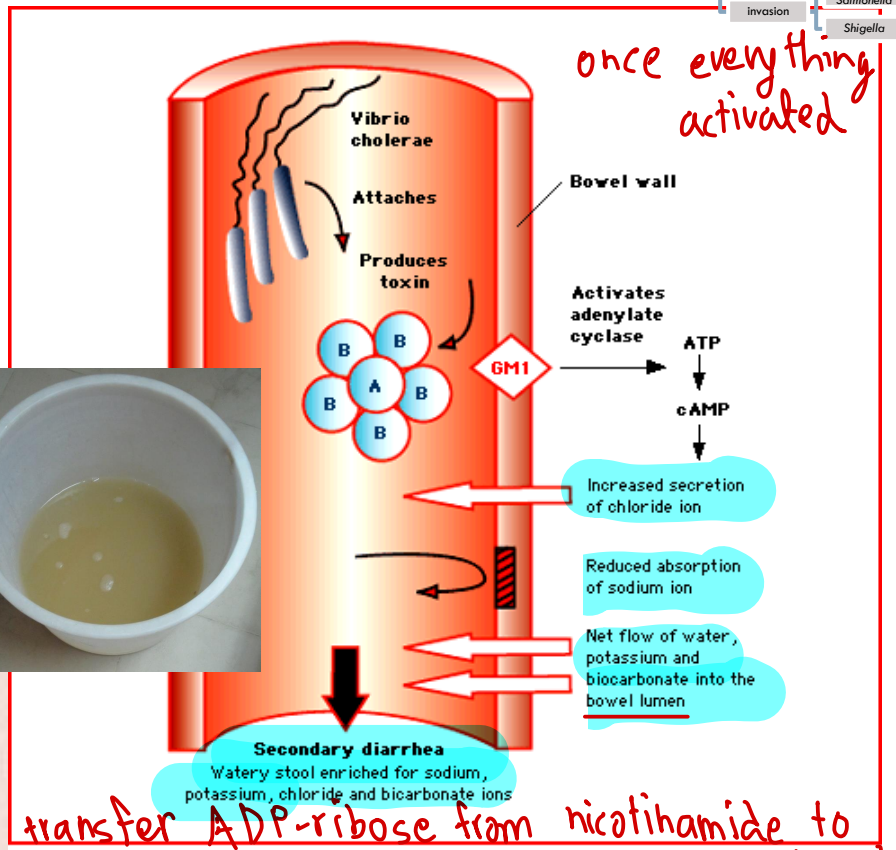


http://lacasamorett.com/foxgallery/vibrio-cholerae-gram-stain.html

Mechanism of cholera toxin









Pathophysiology of Vibrio cholerae infection Schematic representations of the mechanisms involved in diarrhea production by Vibrio cholerae. (Modified from Butterton, JR, Calderwood, SB. Vibrio cholerae 01. In: Infections of the Gastrointestinal Tract, Blaser, MJ, Smith, PD, Ravdin, LI, et al (Eds), Raven Press, New York, 1995, p. 649.)

transfer ADP-ribose from nicotinamide to G protein

A sub unit active G by

activate G protein

NAD (nicotinamide-ADP-ribose) + Gs protein → nicotinamide + G-ADP-ribose

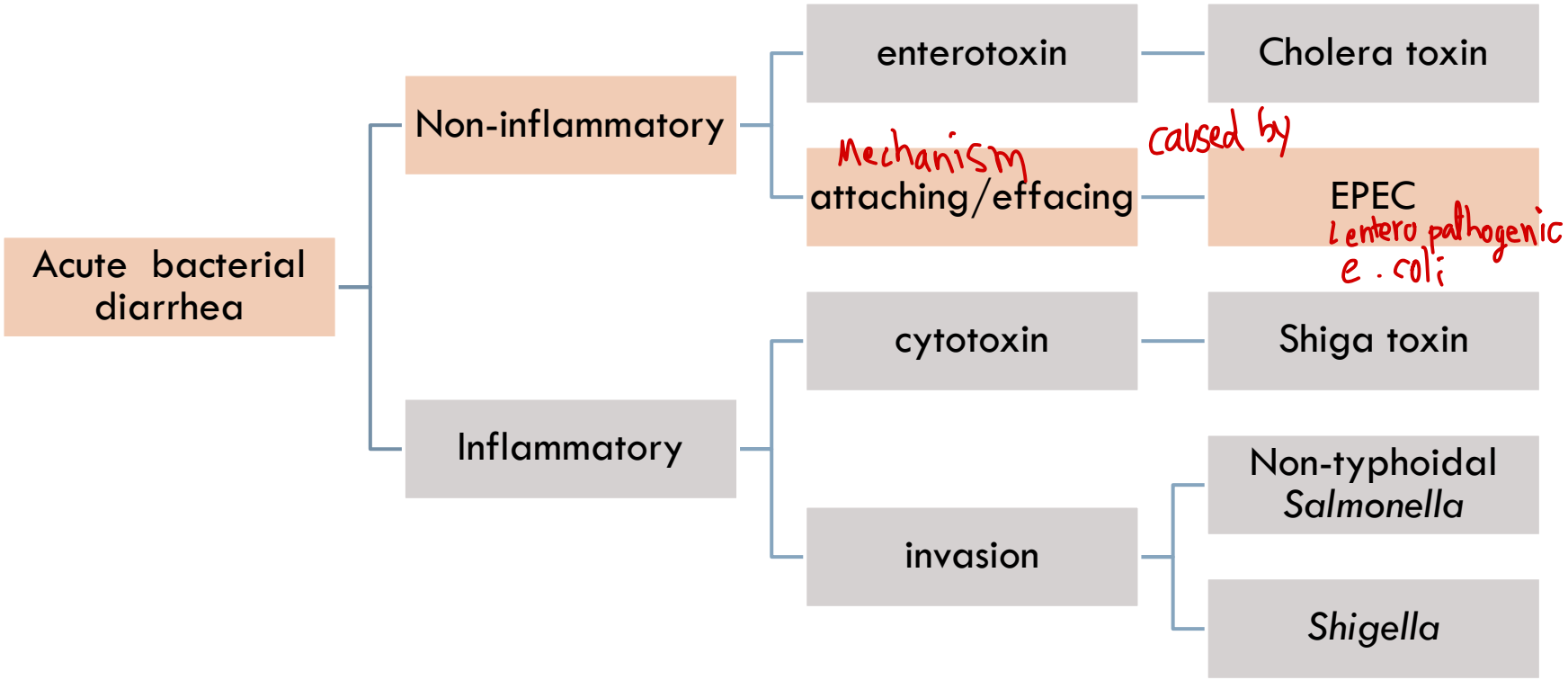
convert → activate cl channel

ATP → cAMP ← Activate adenylate cyclase

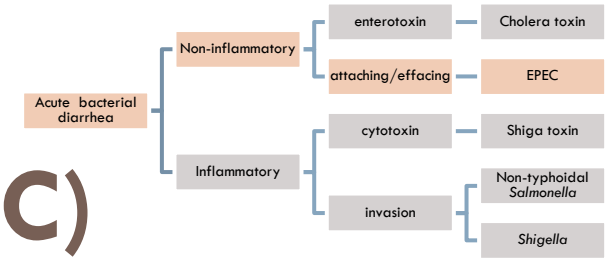
activate

http://www.ebi.ac.uk/interpro/potm/2005_9/Page2.htm

Pathogenesis of acute bacterial diarrhea

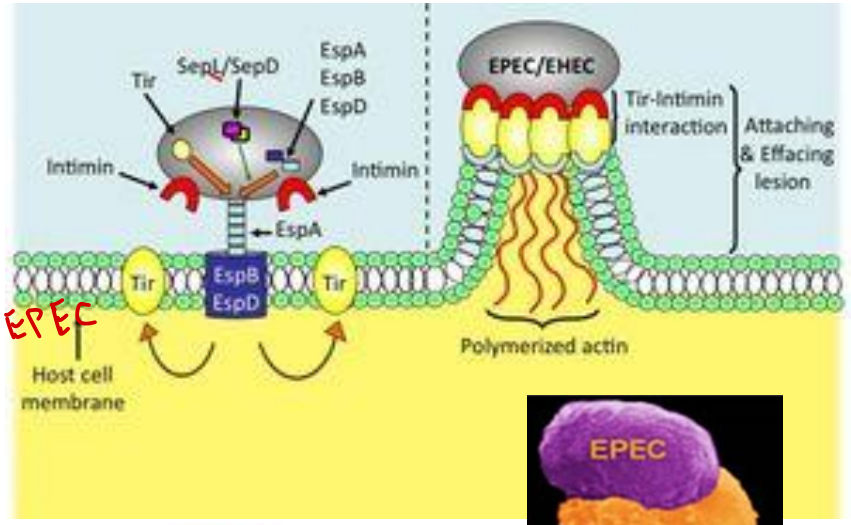


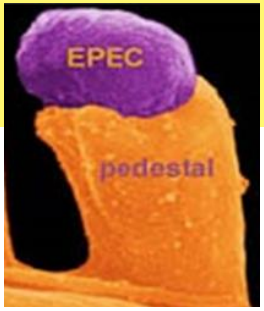
Attachment/effacement: enteropathogenic *E. coli* (EPEC)

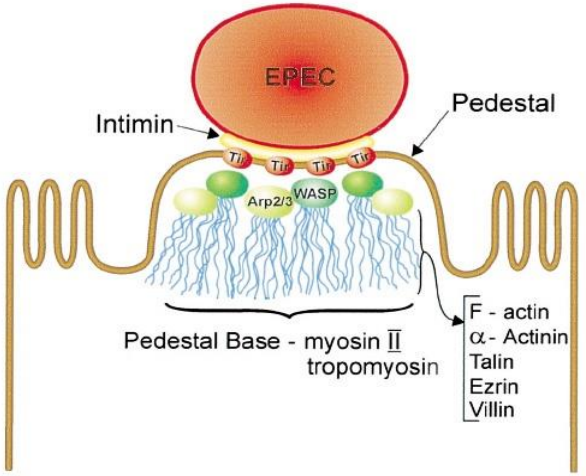


→ protein that injected into host

- **Tir (translocated intimin receptor)**
  - Injected into host cytosol via type III secretion system → from EPEC
  - Pedestal formation at bacterium site
- **A/E lesion of microvilli**
  - Interfere transport of Na^+/Cl^- and cause
  - Watery diarrhea



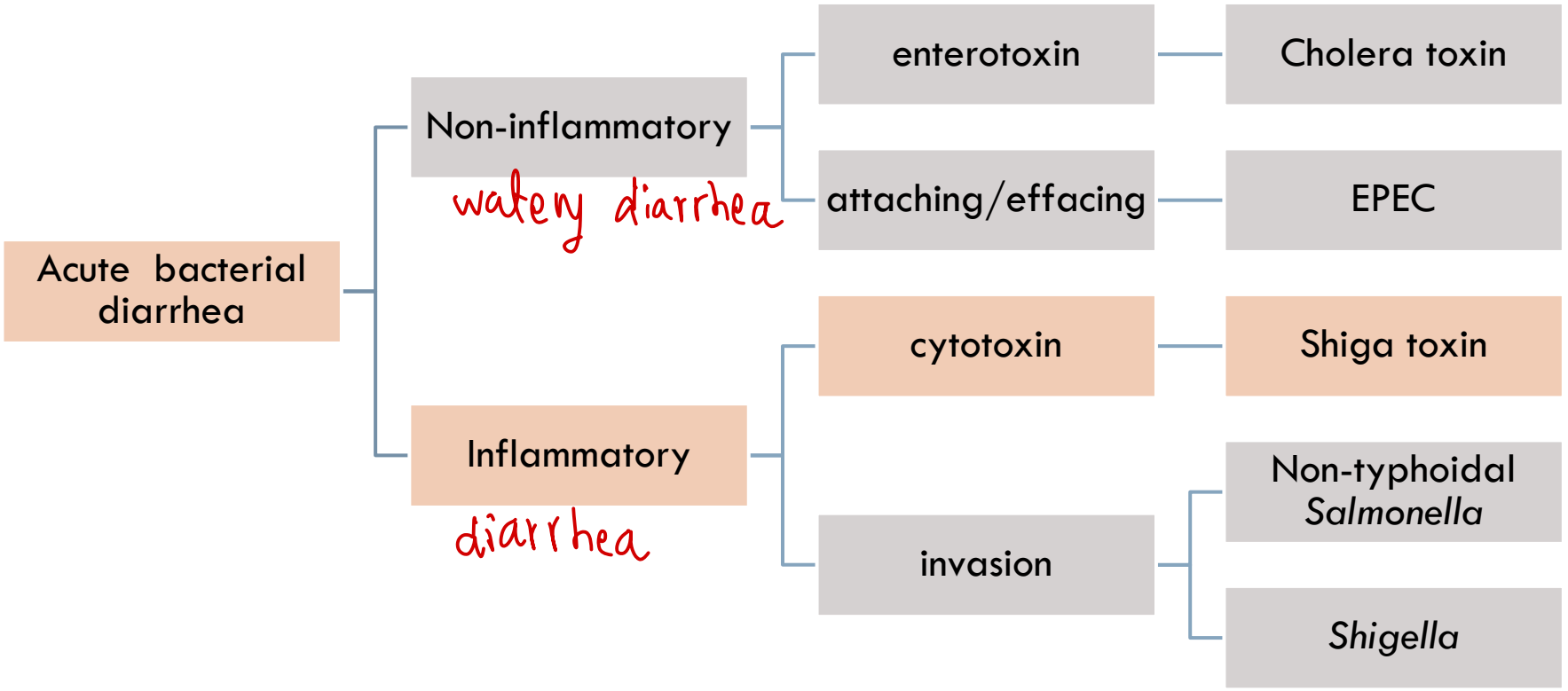




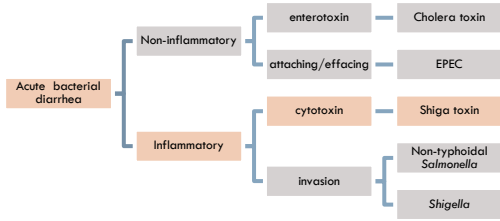


http://www.imperial.ac.uk http://2008.igem.org https://wiki.ubc.ca/Course:PATH4172016W2/Case_4

Pathogenesis of acute bacterial diarrhea



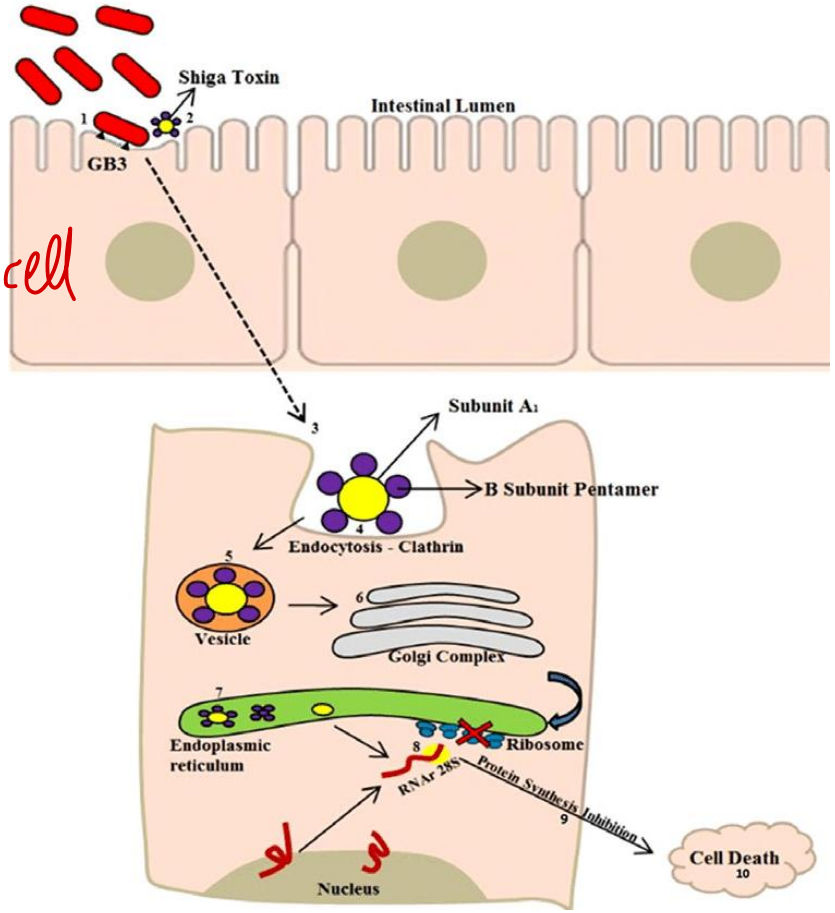
Cytotoxin: Shiga toxin (Stx)



Stx has A, B sub unit

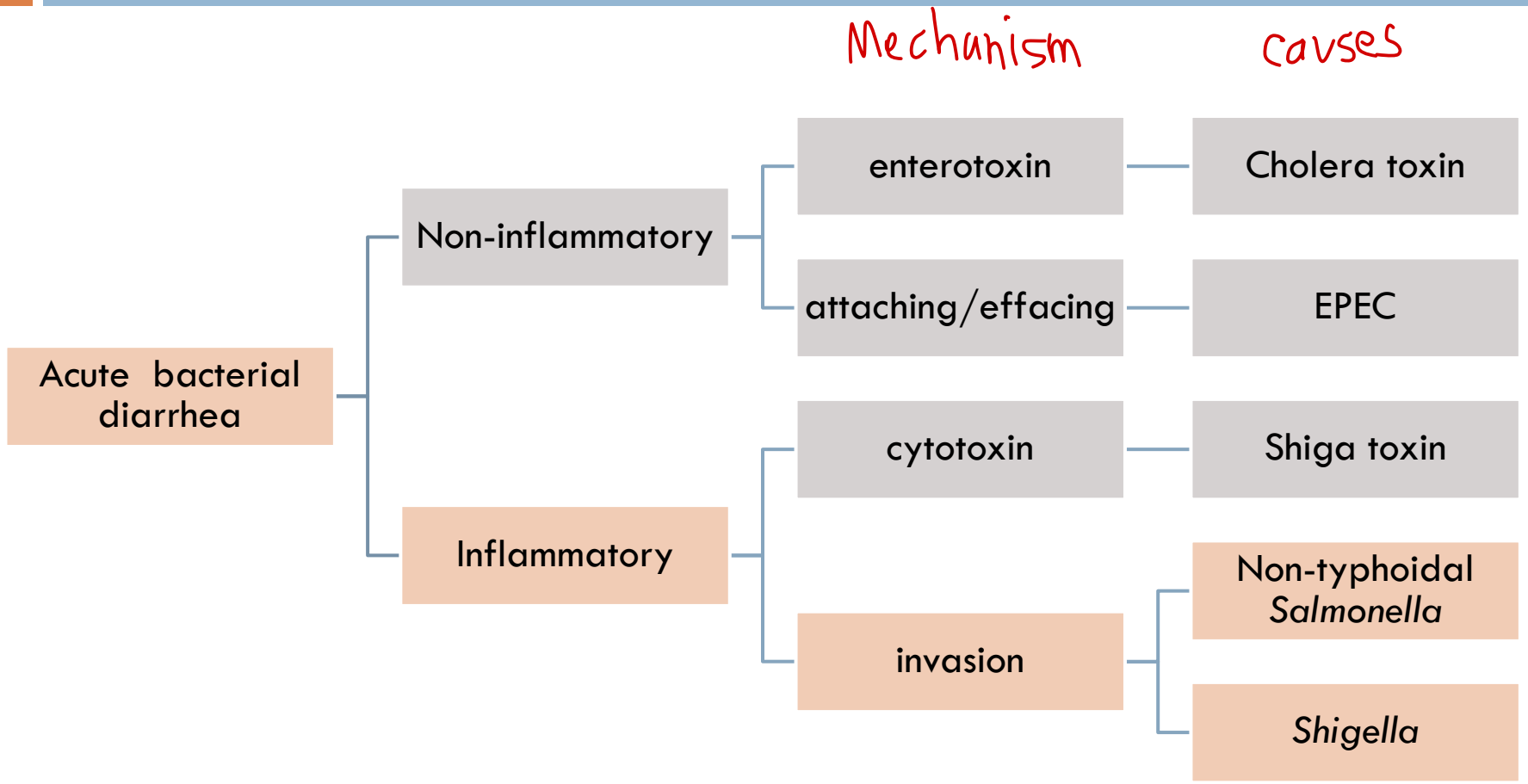
- **B subunit**
  - Bind to host receptor on epithelium
    - globotriaosyl ceremide (Gb3)
- **A subunit** endocytose into epithelial cell
  - Cleave 28S rRNA of 60S ribosomal subunit → inhibit protein synthesis → cell death → mucosal destruction → inflammatory colitis
  - Activates multiple stress signaling and apoptotic pathways → production of inflammatory cytokines by target cells.
- *Shigella dysenteriae* serotype 1, Shiga toxin–producing *Escherichia coli* (STEC)

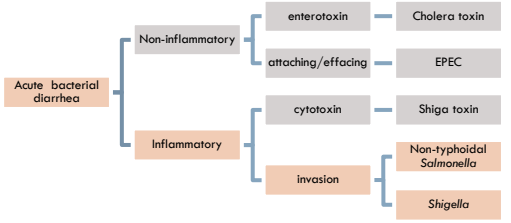
↓ both secrete Stx



C 2017 Institute of Food Technologists® doi: 10.1111/1541-4337.12302

Pathogenesis of acute bacterial diarrhea

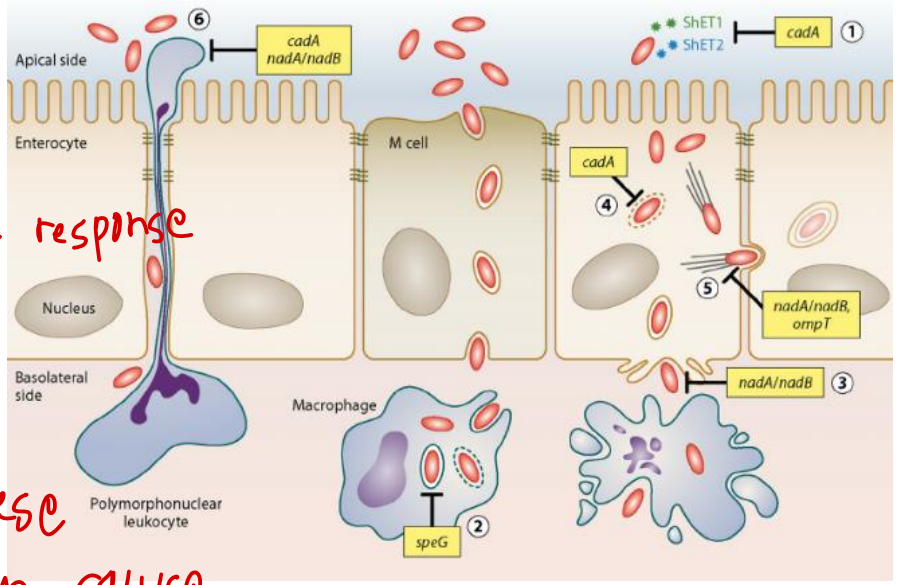




Invasion of epithelial cells

Bacteria can

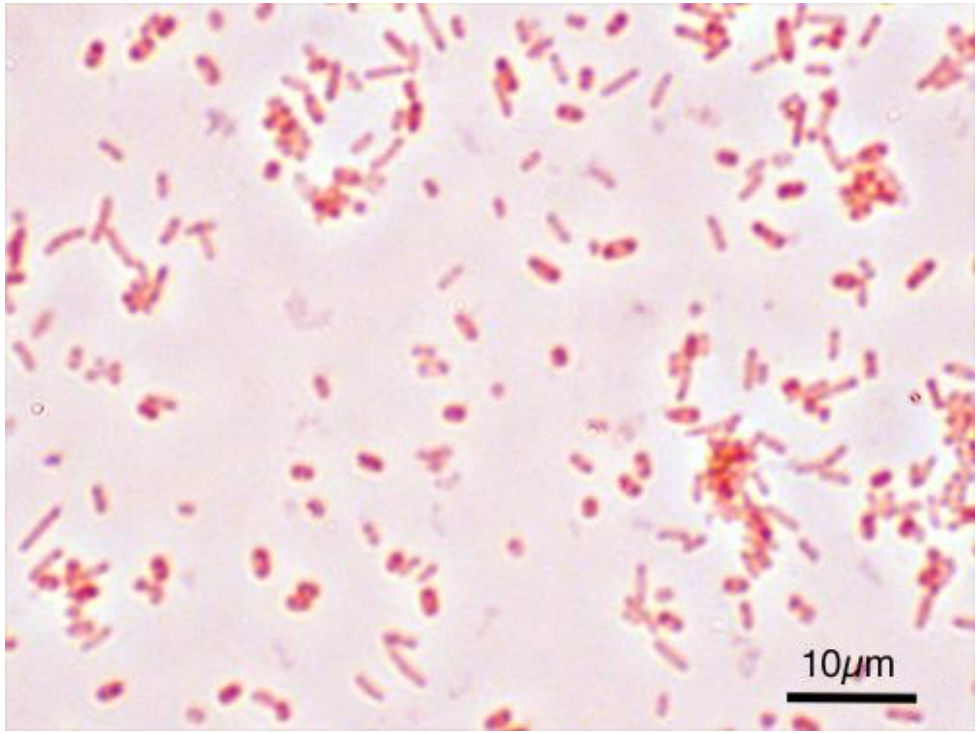
- **Invade and destroy epithelial cells**
- **Induce vigorous host** immune response **and cause inflammatory responses**
  - eg. nontyphoidal *Salmonella*, *Shigella*, Enteroinvasive *E. coli* (EIEC)



These can cause invasion of epithelial cells

• Gram (-) bacili

Salmonella



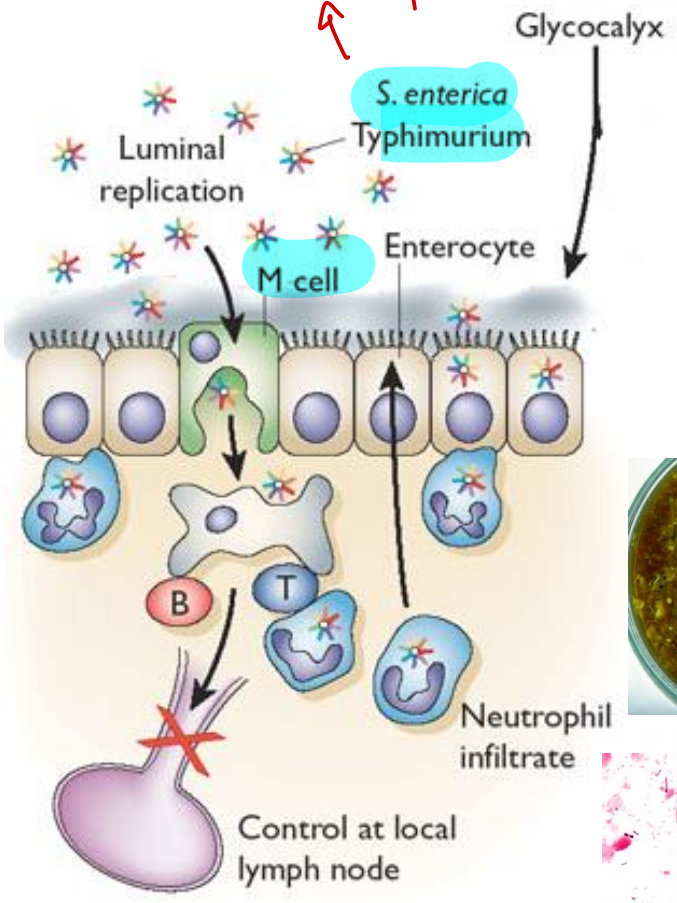
https://commons.wikimedia.org/wiki/File:Salmonella_Typhimurium_Gram.jpg

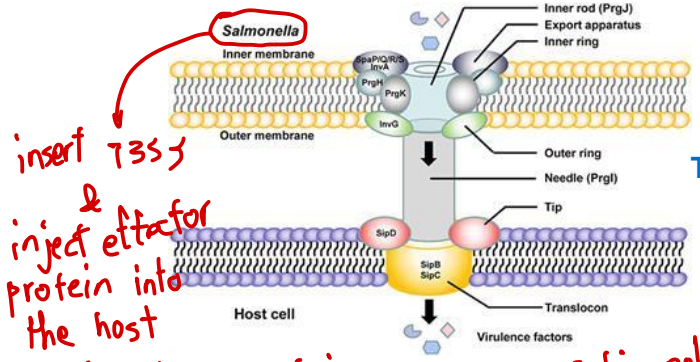
Non-typhoidal *Salmonella*

can be invade by T3SS

can be engulf by M cell

I can cause localized inflammation of the intestinal tract





insert T3SS & inject effector protein into the host

syringe like

Type III secretion system (T3SS)

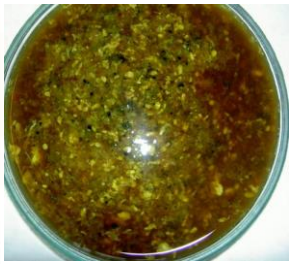
↳ this protein can cause actin polymerization, so epithelial cells that's not usually professional phagocyte can ingest the bac into the cell

↓

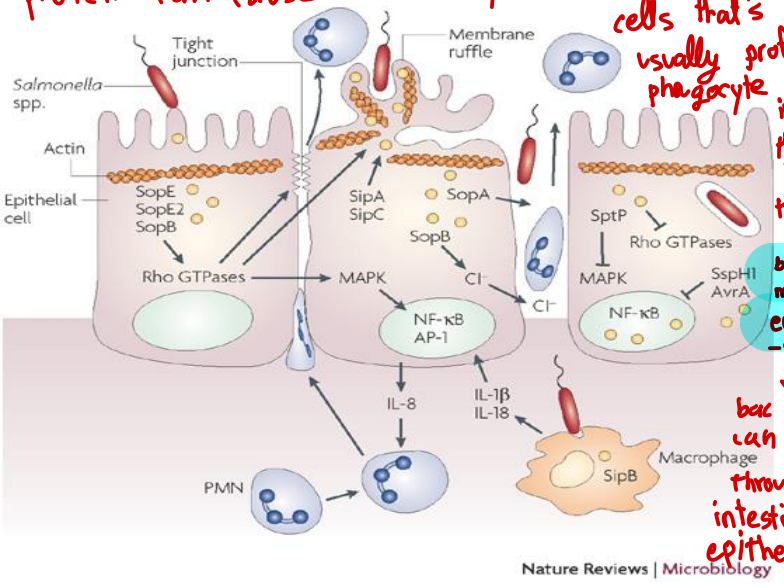
bacterial mediated endocytosis

↓

bac now can invade through intestinal epithelial cell







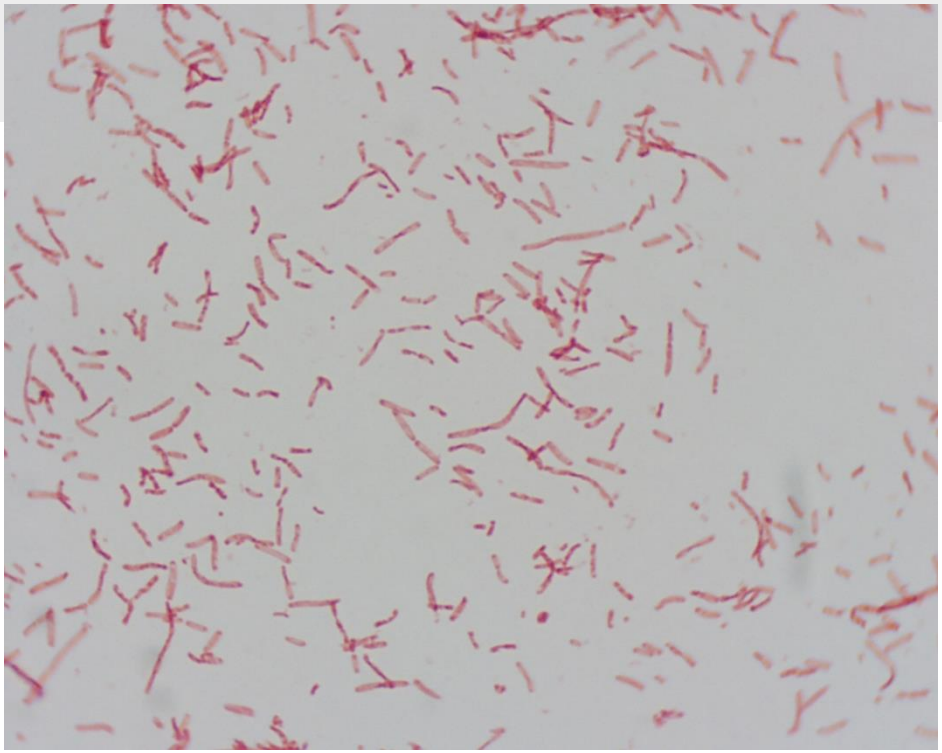
Nature Immunology 2002; 3: 1026-1032

Front. Cell. Infect. Microbiol., 2019 | https://doi.org/10.3389/fcimb.2019.00270

Nat Rev Microbiol 2008; 6: 53-66

Shigella • Gram (-) bacteria

Red Book® Online



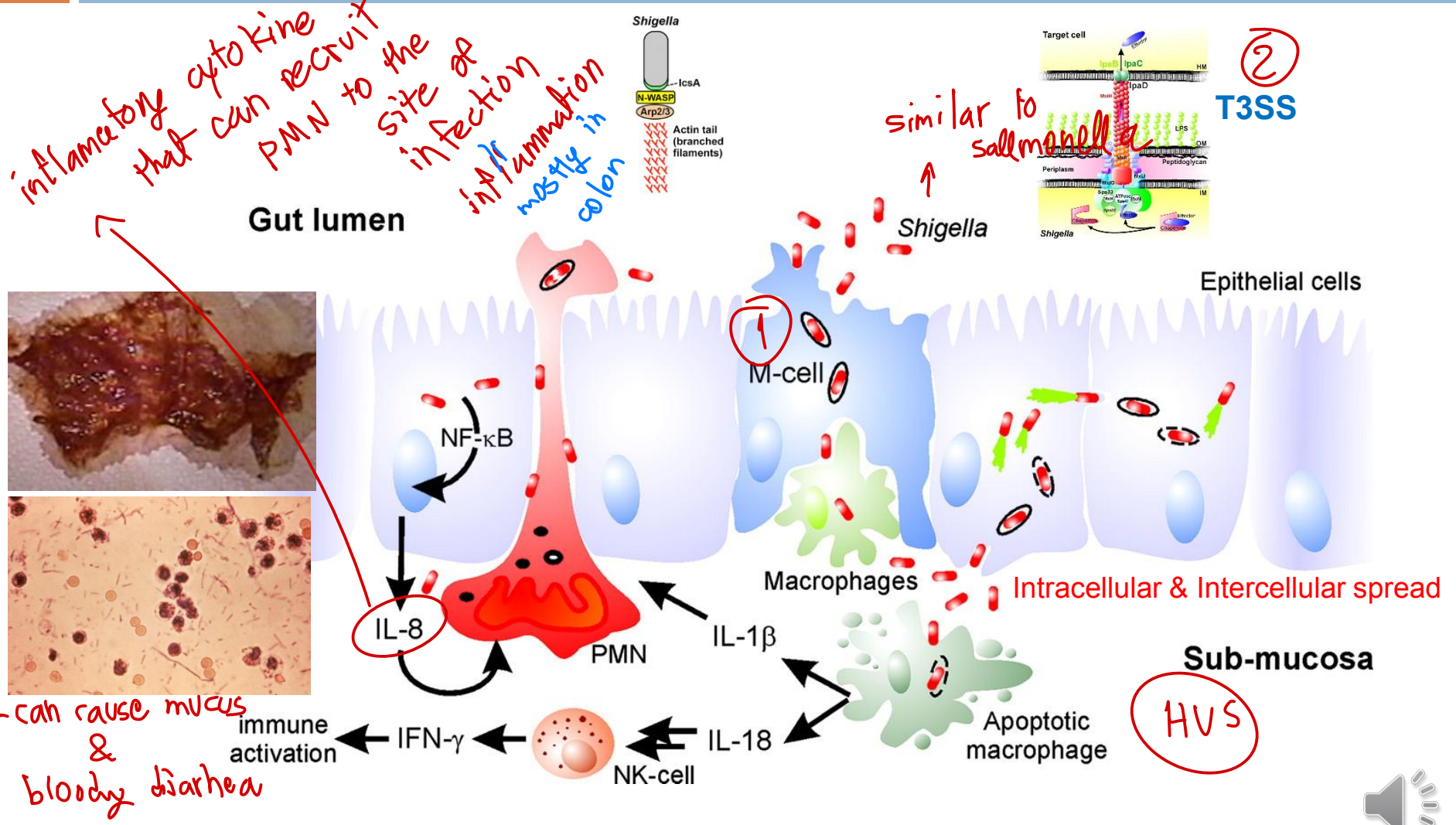
Gram-stain of Shigella sonnei from the previous image (magnification x100). Courtesy of Rita Yee, MT (ASCP) SM

Date of download: 11/28/2016

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Red Book® 2015,

Shigella spp.
Shigellosis (bacillary dysentery)



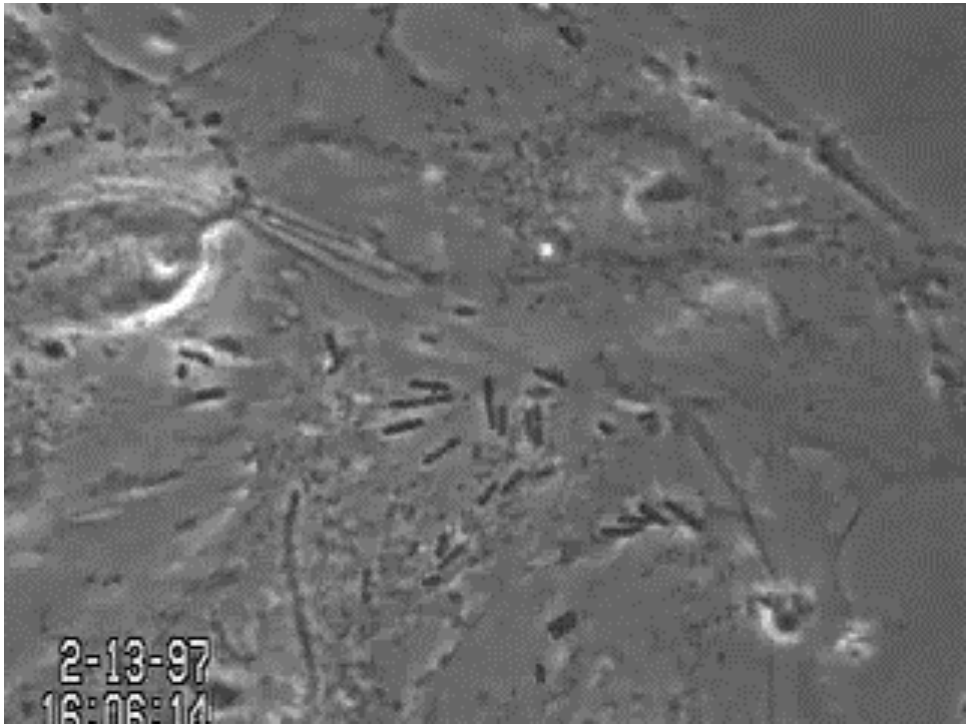
Clinical Microbiology Reviews 2008; 21: 134-156

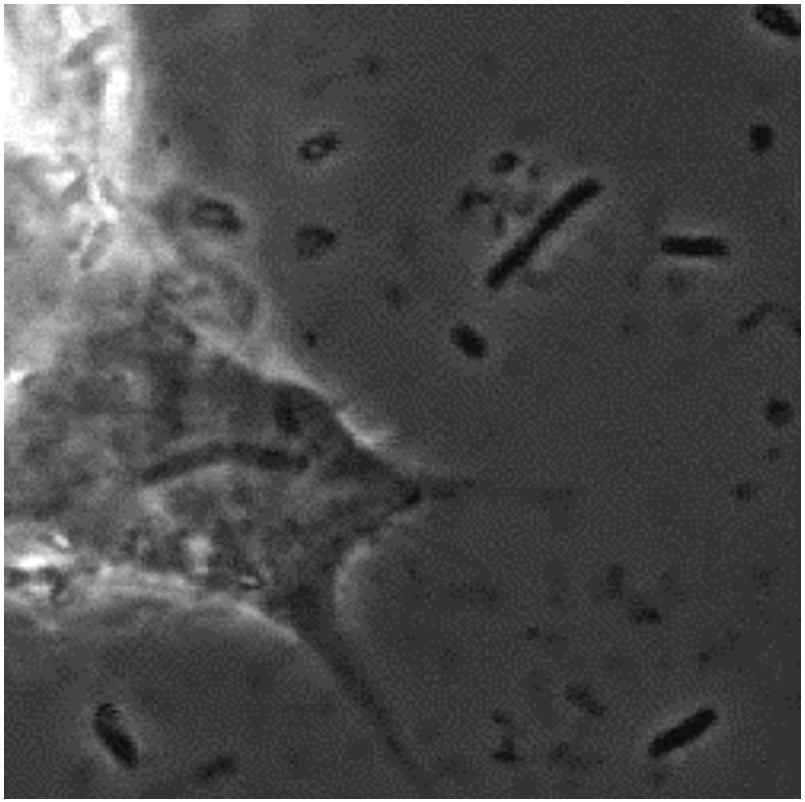
shigella - can secrete toxin by itself
  L if give antibi can induce HUS which can be fetal to the patient

L why antibiotic can induce HUS
  L it's the stress make gene express more of a toxin.

why is the effect of HUS is common in kidney
  L

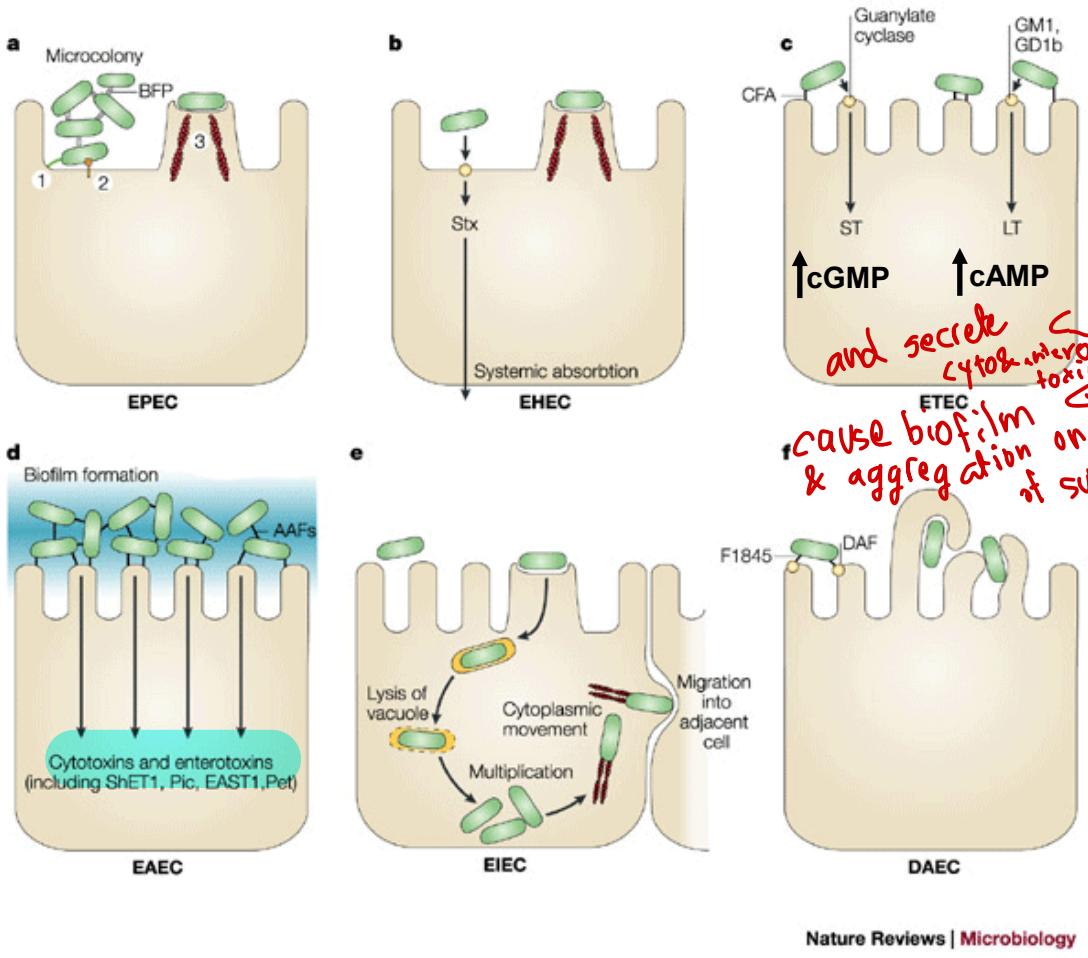
Intracellular and extracellular spread





UpToDate

Diarrheagenic *E. coli*: pathotypes

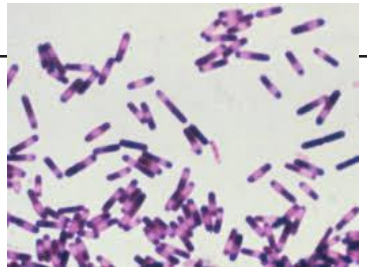


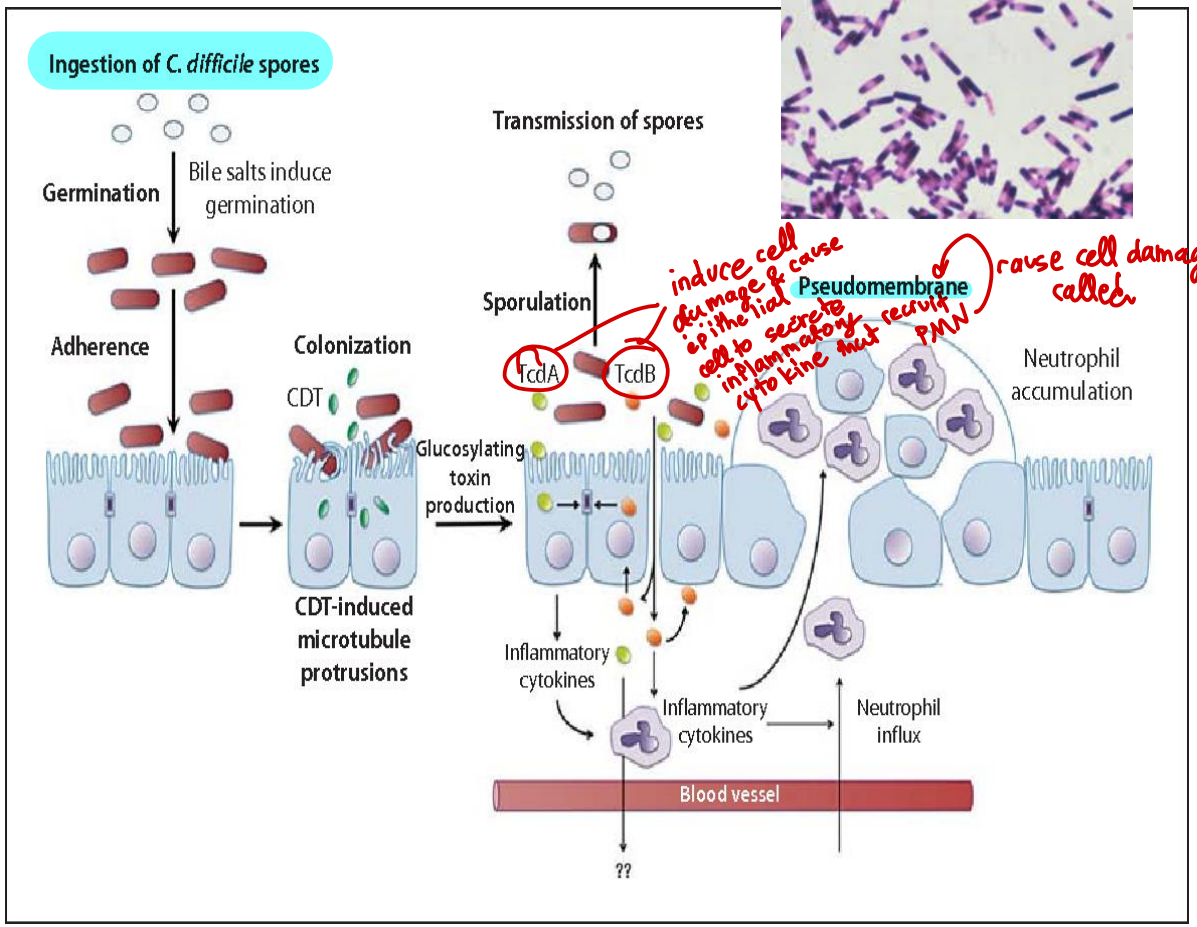
- **Watery diarrhea**
  - EPEC: enteropathogenic
  - ETEC: enterotoxigenic
  - EAEC: enteroaggregative
  - DAEC: diffusely adherent
- **Inflammatory diarrhea**
  - EHEC: enterohemorrhagic [aka Shiga toxin-producing *E. coli* (STEC), verocytotoxin-producing *E. coli* (VTEC)]
  - EIEC: enteroinvasive

Nature Reviews Microbiology volume 2, pages123–140 (2004)

Pathogenesis of antibiotic-associated (pseudomembranous) colitis by *Clostridioides difficile*

aerobic bacteria & spore forming bacteria





induce cell damage & cause epithelial cell to secrete inflammatory cytokine that recruit PMN

cause cell damage called

Pathogenesis of *Clostridium difficile* Infection

Antibiotic Therapy
↓
Alteration of Colonic Microflora
↓
C difficile Exposure and Colonization
↙ ↘
Asymptomatic Colonization
Release of Toxins A and B
↓
CDAD



* can cause
① chronic ulcer
② inflammatory diarrhea

Journal of Innate Immunity 4 (2012): 149 - 158.

Arch Intern Med. 2002;162(19):2177–2184. doi:10.1001/archinte.162.19.2177

Major syndromes of enteric infection



Gastroenteritis

No fever or low grade fever

No white blood cells in stool sample

- *Vibrio cholerae*
- ETEC
- EAggEC
- EPEC

→ Secretory diarrhoea

Fever

White blood cells in stool sample (neutrophils)

- *Campylobacter jejuni*
- *Shigella* spp.
- Non-typhoidal *Salmonella* serotypes
- EIEC
- EHEC

→ Inflammatory diarrhoea



Febrile patient

Fever

White blood cells in stool sample (mononuclear cells)

- Typhoidal *Salmonella* serotypes
- Enteropathogenic *Yersinia* spp.
- *Brucella* spp.

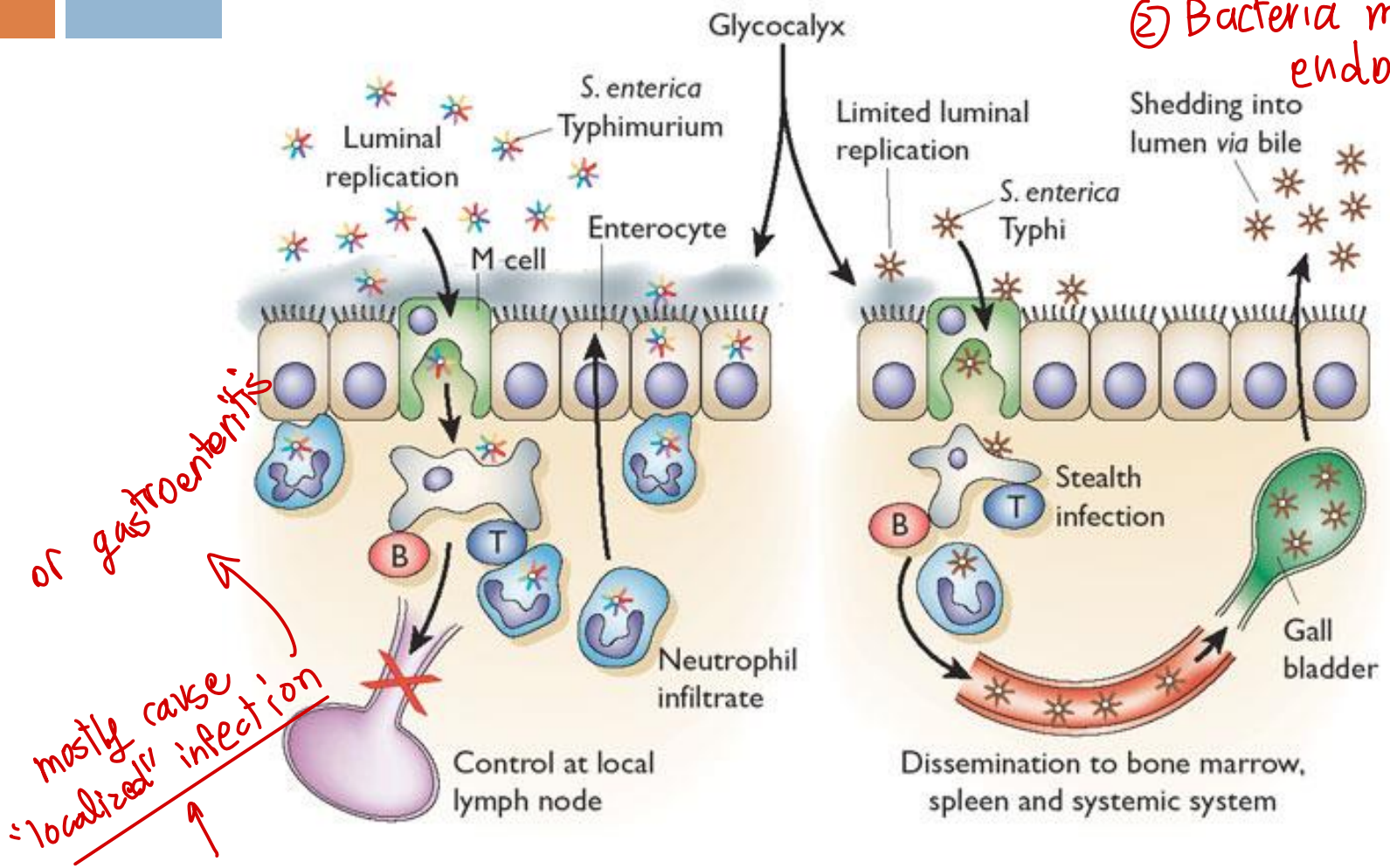
→ Enteric fever

does not present with diarrhea, but present with fever

Nat Rev Microbiol 2008

Salmonella spp.

① M cell
② Bacteria mediated endocytosis (TTSS)



or gastroenteritis

mostly cause "localized" infection

Non-typhoidal *Salmonella* Gastroenteritis

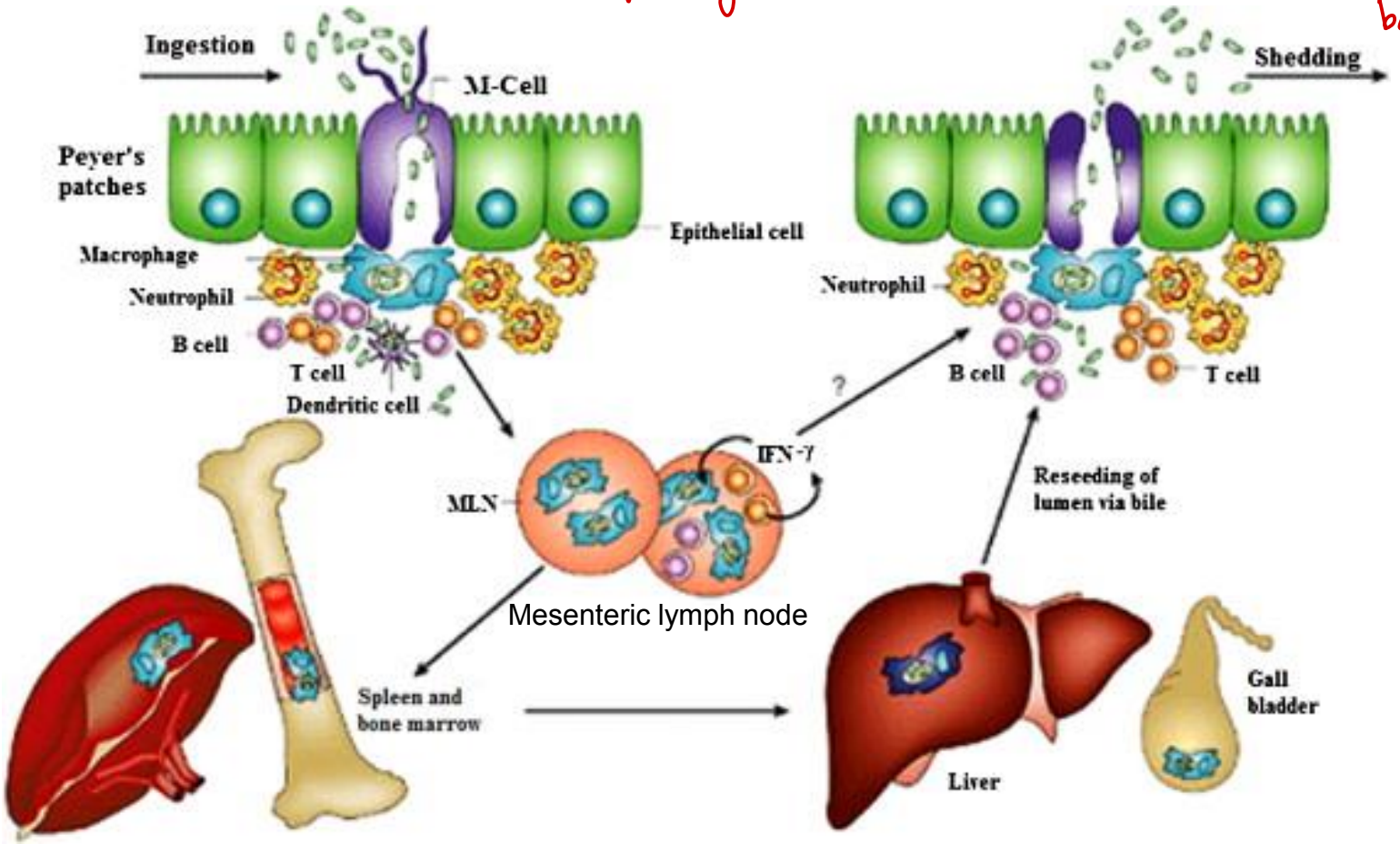
Salmonella Typhi Enteric fever (systemic infection)

Enteric fever (typhoid fever): *Salmonella enterica* serotype Typhi

cause by serotype not species

can survive in macrophage

base on antigen of bacteria

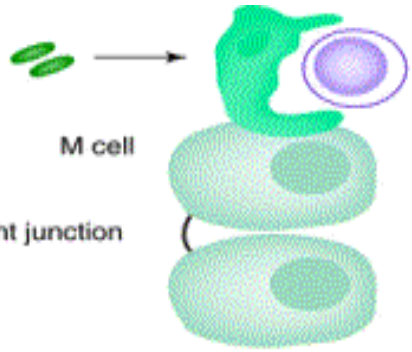


Microbiological Research 167 (2012) 199– 210

Incubation period 7-14 d

1st wk: bacteremia fever, chills

Salmonella Typhi



Macrophage

First exposure of Peyer's patch to *S. typhi*

Mesenteric lymph node

Blood

Spleen



Bone marrow



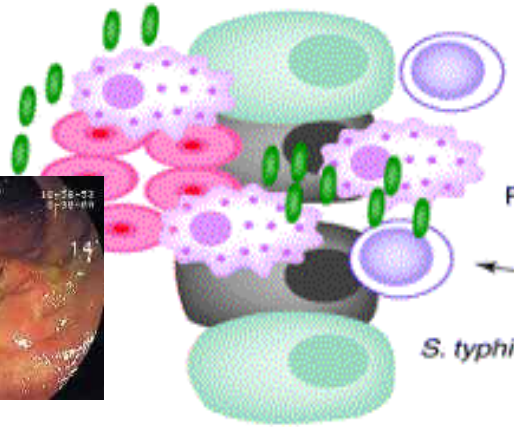
skin

Blood

2nd wk: rose spot, abdominal pain

Liver

Via gall bladder





2nd exposure of Peyer's patch to *S. typhi*

3rd wk: hepatosplenomegaly, Intestinal perforation and bleeding, secondary bacteremia, peritonitis, sepsis

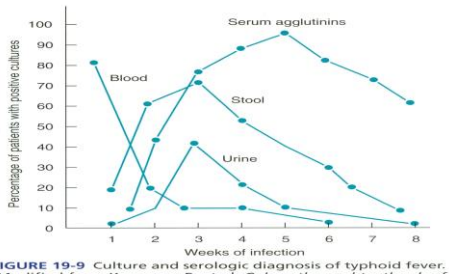
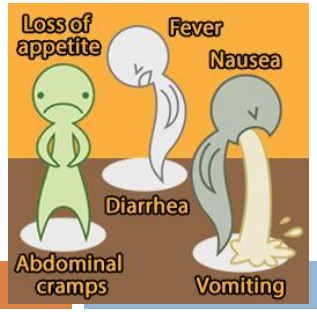


FIGURE 19-9 Culture and serologic diagnosis of typhoid fever. (Modified from Koneman E, et al: *Color atlas and textbook of diagnostic microbiology*, ed 4, Philadelphia, 1992, Lippincott.)

Key:

- Peyer's patch
- *Salmonella typhi*
- Macrophage
- Lymph node
- Red blood cells
- Necrotic Peyer's patch
- T cells

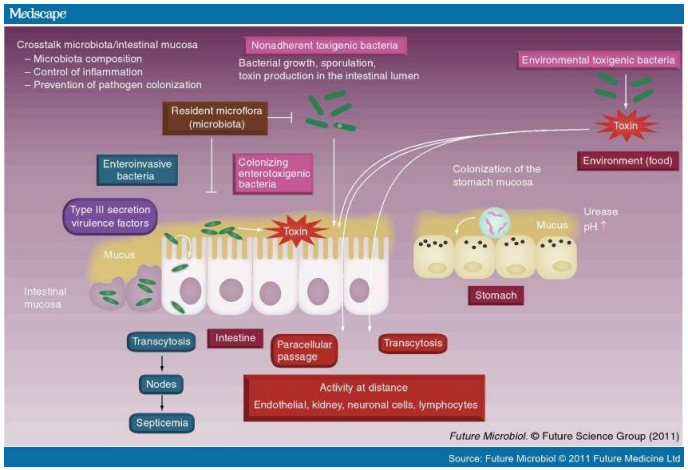
TRENDS in Microbiology



Food poisoning

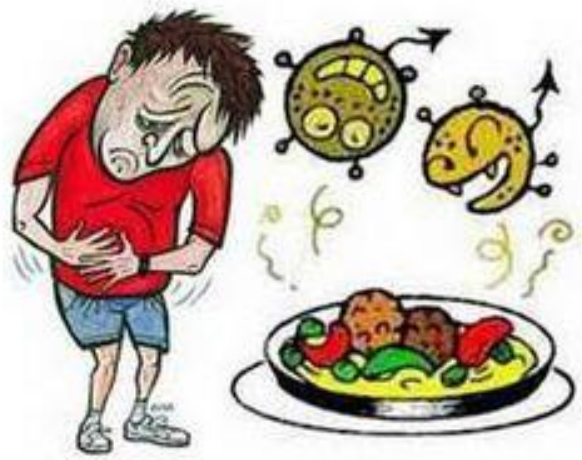


- Food poisoning: illness caused by the consumption of food or water contaminated with microbes and/or their toxins
- Outbreak: two or more cases of a similar illness that happens after people eat a common food
- Incubation period: mechanisms of disease (depend on ↴)
  - Preformed toxins (can cause by one of these)
  - Toxin-producing bacteria
  - Invasive bacteria



Food poisoning: common food sources

- **Poultry**
  - *Salmonella, Campylobacter, Shigella*
- **Beef, unpasteurized fruit juice**
  - Enterohemorrhagic *E. coli* (EHEC)
- **Seafood**
  - *Vibrio parahaemolyticus, Vibrio vulnificus*
- **Cheese, milk**
  - *Listeria*
- **Egg**
  - *Salmonella*
- **Fried rice**
  - *Bacillus cereus*



Incubation Period/ Organisms (catagorize base on incubation period)	Symptoms	Common Food Sources
1 - 6 h **Preformed toxin**		
Staphylococcus aureus heat-stable enterotoxin	Nausea, vomiting, diarrhea	Ham, poultry, potato or egg salad, mayonnaise, cream pastries
Bacillus cereus heat-stable enterotoxin, short-incubation or emetic form	Nausea, vomiting, diarrhea	Fried rice
8 - 16 h		
Clostridium perfringens heat-labile enterotoxin	Abdominal cramps, diarrhea (vomiting rare)	Beef, poultry, legumes, gravies
B. cereus heat-labile enterotoxin, long-incubation or diarrheal form	Abdominal cramps, diarrhea (vomiting rare)	Meats, vegetables, dried beans, cereals
> 16 h		
Vibrio cholerae	Watery diarrhea	Shellfish
Enterotoxigenic *Escherichia coli*	Watery diarrhea	Salads, cheese, meats, water
Enterohemorrhagic *E. coli*	Bloody diarrhea	Ground beef, roast beef, salami, raw milk, raw vegetables, apple juice
Salmonella spp.	Inflammatory diarrhea	Beef, poultry, eggs, dairy products
Campylobacter jejuni	Inflammatory diarrhea	Poultry, raw milk
Shigella spp.	Dysentery	Potato or egg salad, lettuce, raw vegetables
Vibrio parahaemolyticus	Dysentery	Mollusks, crustaceans

Bacillus cereus

what is preform of this.



gram (+) bacili
aerobic

Gastroenteritis (Food Poisoning)

2 type of toxin

Emetic toxin (cereulide)
= Preformed toxin

① Emetic

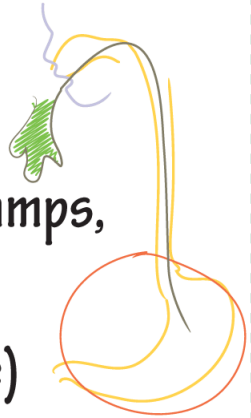
can irritated vagus n.

Heat-stable toxin
Spores survive initial cooking;
Germinate if not refrigerated.
Reheating does NOT kill toxin.

withstand 121 °C for 90 min

Rice





Nausea, abdominal cramps, vomiting.

Complication:
Liver failure (rare)

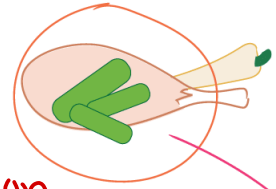
Quick:
Incubation < 6 hrs;
Duration < 24 hrs

② Diarrheal

Heat-labile enterotoxin
Intestinal epithelial cells → Increases [cAMP]

Inactivated after heating at 56 °C for 5 min

Meat, vegetables



↑ after ingestion
Bacteria multiplies in GI tract.



Nausea, abdominal cramps, watery diarrhea

Longer:
Incubation > 6 hrs;
Duration > 24 hrs

- stimulate afferent vagal N in the intestinal mucosa and activate vomiting center in the brain

https://www.drawittoknowit.com/

Complications of bacterial enteric infection

Complication	Important Bacterial Agents	Clinical Considerations
Dehydration	*Vibrio cholerae*, any bacterial enteropathogen	Most important complication of all forms of acute watery diarrhea; should prompt aggressive fluid and electrolyte replacement, usually in hospital
Bacteremia	Salmonella, *Campylobacter fetus*	Organisms that deeply penetrate the intestinal mucosa are prone to cause bacteremia; certain high-risk conditions predispose to systemic salmonella infection
Hemolytic–uremic syndrome	Shiga toxin–producing *Escherichia coli*	Shiga toxin is absorbed, causing injury to endothelial cells of the glomerular capillaries with intravascular coagulation
Guillain–Barré syndrome	*Campylobacter jejuni*	Most cases occur as a result of molecular mimicry, with antibodies directed to campylobacter lipooligosaccharides and peripheral-nerve gangliosides; probability of development of Guillain–Barré syndrome within 2 mo after campylobacter infection estimated at <2/10,000 cases[40]

nerve injury & neurological symptom

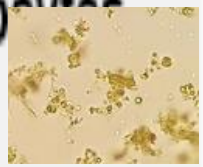
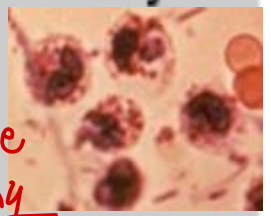
Complications of bacterial enteric infection

Complication	Important Bacterial Agents	Clinical Considerations
Reactive arthritis and iritis	Campylobacter, salmonella, *Shigella flexneri*, yersinia	Occurs in 2.1/100,000 cases of campylobacter infection and 1.4/100,000 cases of salmonella infection; affected persons may be HLA-B27–positive or HLA-B27–negative[41]
Postinfectious irritable bowel syndrome	Inflammatory bacterial pathogens (e.g., campylobacter) are most important, but most bacterial pathogens can produce the syndrome	Enteric bacterial infection with intestinal inflammation in a susceptible host leads to altered intestinal findings and postinfectious irritable bowel syndrome[42-44]; duration is ≥5 yr[45-47]



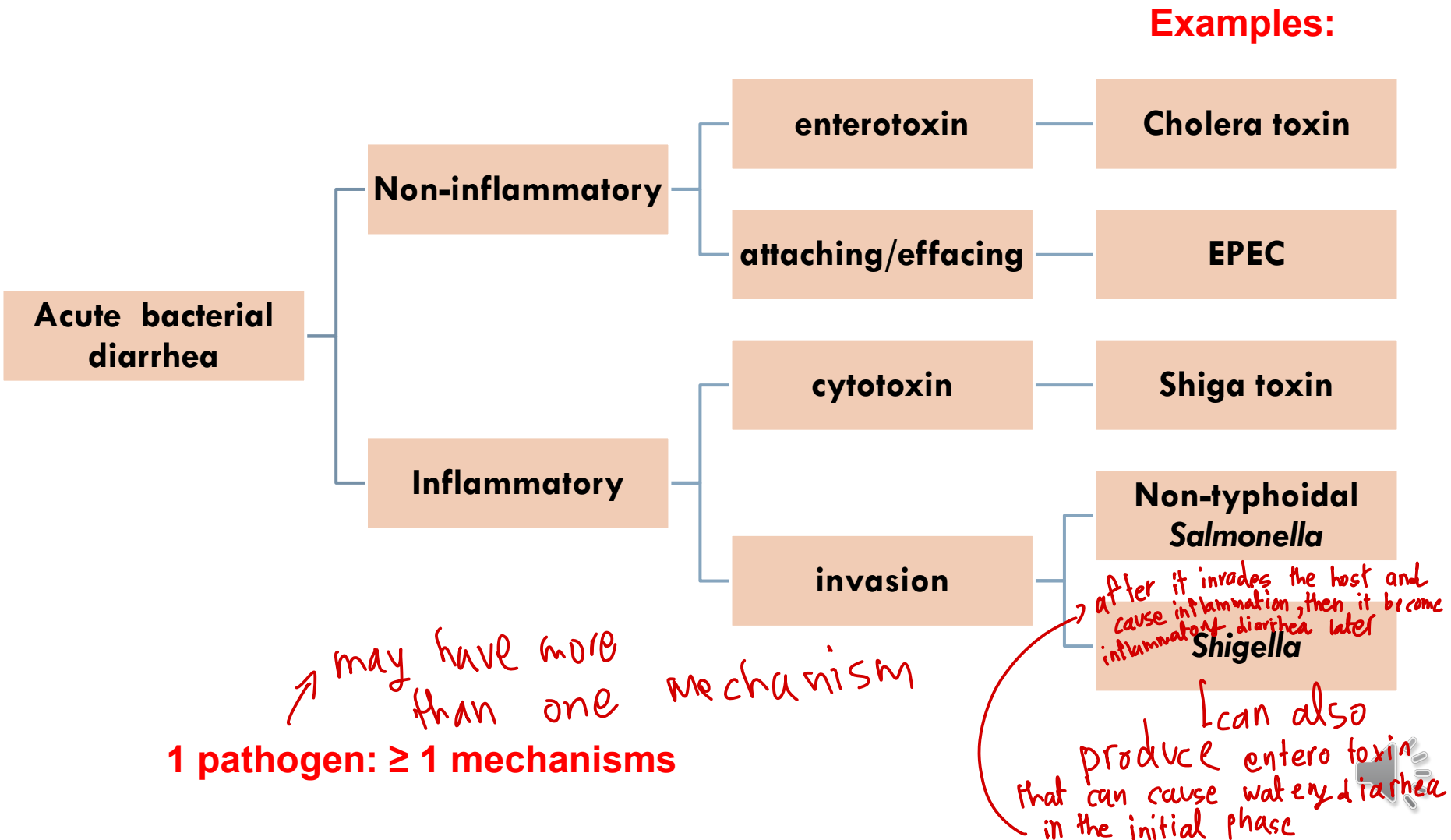
Summary

enteric Bacterial infection of

Mechanism	Location	Illness	Stool Findings
Noninflammatory (enterotoxin)	Proximal small bowel	Watery diarrhea (secretory)	No fecal leuko 
Inflammatory (invasion or cytotoxin)	Colon or distal small bowel	Dysentery or inflammatory diarrhea	Fecal PMN leukocytes +/- RBC  [sometime with bleeding if there is diarrhea]
Penetrating	Distal small bowel, bacteria survive in macrophage and migrate to mesentaric lymph node	Enteric fever	Fecal mononuclear leukocytes 

Modified from Harrison's Principle of Internal Medicine 16th ed

Pathogenesis of acute bacterial diarrhea



Summary

Mechanism	Location	Illness	Stool Findings	Examples of Pathogens Involved
Noninflammatory (enterotoxin)	Proximal small bowel	Watery diarrhea	No fecal leukocytes; mild or no increase in fecal lactoferrin	*Vibrio cholerae*, enterotoxigenic *Escherichia coli* (LT and/or ST), entero-aggregative *E. coli*, *Clostridium perfringens*, *Bacillus cereus*, *Staphylococcus aureus*, *Aeromonas hydrophila*, *Plesiomonas shigelloides*, rotavirus, noro-virus, enteric adenoviruses, *Giardia lamblia*, *Cryptosporidium* spp., *Cyclospora* spp., microsporidia
Inflammatory (invasion or cytotoxin)	Colon or distal small bowel	Dysentery or inflammatory diarrhea	Fecal polymorphonuclear leukocytes; substantial increase in fecal lactoferrin	*Shigella* spp., *Salmonella* spp., *Campylobacter jejuni*, enterohemorrhagic *E. coli*, enteroinvasive *E. coli*, *Yersinia enterocolitica*, *Listeria monocytogenes*, *Vibrio parahaemolyticus*, *Clostridium difficile*, *A. hydrophila*, *P. shigelloides*, *Entamoeba histolytica*, *Klebsiella oxytoca*
Penetrating	Distal small bowel	Enteric fever	Fecal mononuclear leukocytes	*Salmonella typhi*, *Y. enterocolitica*

Abbreviations: LT, heat-labile enterotoxin; ST, heat-stable enterotoxin.

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Q&A

kpatarakul@gmail.com